LiDAR-Augmented Wi-Fi Signal Strength Prediction: A Use-Case Study on Industrial AGVs

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***Abstract*—Industrial automated guided vehicles (AGVs) require reliable Wi-Fi connectivity for safety telemetry, traffic management, mission updates, and remote intervention. However, signal strength in industrial workspaces is hard to predict from the access point (AP) distance alone due to waveguiding, blockage, and multipath. Since industrial AGVs are equipped with safety LiDARs that capture environmental geometry, the natural hypothesis is that LiDAR-derived features can improve Wi-Fi signal strength prediction over what AP-relative geometry alone provides. This paper presents a use-case study of LiDAR-augmented signal strength prediction on data gathered from a deployed industrial AGV (three routes, 681,593 matched samples). Two evaluation protocols are used: cross-session and within-session generalization. Five robustness controls are applied: an XGBoost hyperparameter sweep, a LightGBM framework cross-check, a 1-meter spatial buffer-zone exclusion, a depth-constrained overfitting check, and LiDAR-feature permutation tests. Across the configurations tested, ego-frame LiDAR features did not contribute robust value over a 10-feature baseline of position, telemetry, and AP-relative geometry; the only nominal positive was diagnosed by the permutation control as marginal-distribution-driven rather than row-aligned LiDAR signal. The lowest-RMSE LiDAR-free model achieves an RMSE of 3.96 dB on `signal_power`, approximately 1 dB below the dataset's noise floor. The paper presents the first empirical study of LiDAR-augmented Wi-Fi signal strength prediction for industrial AGVs and identifies a minimal feature set sufficient for sub-noise-floor within-session prediction in the studied scenario.**

***Index Terms*—industrial AGV, Wi-Fi link quality, LiDAR, signal power prediction, XGBoost, SHAP, eXplainable AI (XAI)**

I. Introduction

With the ongoing adoption of Industry 4.0 and 5.0 paradigms, automated guided vehicles (AGVs) are increasingly deployed in warehouse, manufacturing, and distribution environments [1]. Reliable wireless communication supports the coordination functions of AGV fleets, including safety telemetry, traffic management, mission updates, and remote intervention. Thus, unpredictable signal quality can result in idle time, near-collisions, or unplanned safety stops. Wi-Fi remains the dominant choice for these links due to its cost, infrastructure availability, and coverage characteristics, but the conditions under which industrial Wi-Fi performs reliably differ from those of office or residential installations [2], [3]. Metallic shelving, structural columns, loading equipment, and moving forklifts produce a propagation environment dominated by waveguiding, blockage, and multipath. Empirical log-distance models, even with precisely measured access point (AP) coordinates, fit indoor industrial measurements with considerably more residual variance than outdoor or campus deployments [2], [4]. Field-deployable predictors of received signal strength along a planned AGV trajectory are therefore operationally valuable: they let the fleet coordination software pre-compute coverage, flag risk zones, and schedule jobs around predicted dropout regions [5].

Industrial AGVs compliant with ISO 3691-4 [6] are typically equipped with 2D LiDARs for personnel detection, protective field monitoring, and on-board localization. These scanners continuously produce a wide-angle range scan of the AGV's local environment, capturing corridor geometry, blockage along arbitrary directions, and clutter density. Indoor Wi-Fi propagation is primarily shaped by these geometric properties, which motivates the question of whether LiDAR-derived environmental features can improve signal strength prediction beyond what AP-relative geometry alone provides, and whether the added information justifies the integration effort. Recent work has shown that point-cloud-derived environmental features can improve mmWave-band channel prediction [7]–[10]. Existing AGV Wi-Fi link quality predictors [4], [11], [12] use telemetry alone and do not incorporate environmental geometry. To our knowledge, the intersection of LiDAR-derived environmental features, Wi-Fi link quality prediction, and industrial AGV deployment has not been explored.

This paper presents a use-case study, conducted on data gathered from a deployed industrial AGV across three routes (one *session* per route), that examines whether ego-frame LiDAR features improve over a baseline that uses only AGV position, telemetry, and AP-relative geometry. The paper addresses the following research questions (RQs):

- **RQ1.** Do LiDAR-derived environment features improve received signal strength prediction over a baseline of AGV position, telemetry, and AP-relative geometry, in cross-session generalization?
- **RQ2.** Do LiDAR-derived environment features improve prediction in within-session leave-region-out (LRO) generalization (train/test on disjoint sub-regions of one route), after controls for spatial autocorrelation and tree-depth overfitting?
- **RQ3.** Are the observed effects robust to feature group substitution, hyperparameter and framework choice, and to a LiDAR-feature permutation control?

The contributions of this paper are summarized as follows:

- The first controlled empirical evaluation of LiDAR-augmented Wi-Fi signal strength prediction on data gathered from a deployed industrial AGV is performed.
- A feature group ablation procedure is performed that separates the case where a feature group provides no useful information from the case where it is substitutable for another retained group.
- A minimal feature set is identified that is sufficient for accurate within-session signal strength prediction in the studied scenario.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the dataset, its preparation, and the feature stack. Section IV presents the experimental methodology. Sections V and VI report the cross-session and within-session experiments, respectively. Section VII presents the feature attribution analysis. Section VIII discusses the implications and limitations, and Section IX concludes. The dataset, source code, analysis scripts, and result reports are released to support reproduction and follow-up work[1].

II. Related Work

A coherent body of work on industrial Wi-Fi link quality prediction has used telemetry-only features without explicitly modeling the surrounding workspace. Linear combinations of exponential moving averages [13] and shallow neural networks [12], [14] reach competitive accuracy on warehouse-deployed equipment, and recent extensions replace shallow models with deep architectures [4], [11]; the feature inventory across these works remains essentially telemetric. Broader surveys on machine learning for IEEE 802.11 [15], [16] catalog similar feature sets and identify environmental modeling as an under-investigated direction, particularly in industrial deployments where waveguiding and blockage dominate. The fundamental challenges of Wi-Fi reliability for moving equipment in such environments are also documented in [2]. The cumulative picture is that telemetry features alone leave residual variance that prior state-of-the-art does not account for, motivating the search for additional features that capture environmental structure.

The use of geometric environmental features for radio-channel prediction has been explored most extensively in the millimeter-wave (mmWave) band. The work [7] uses raw LiDAR point clouds to anticipate line-of-sight blockage by pedestrians in 60 GHz wireless LAN, demonstrating that point cloud-based prediction is feasible at indoor scales without the privacy concerns of camera-based methods. The authors of [8] fuse LiDAR scans with panoramic imagery in a PointNet-based regressor for prediction of mmWave channel parameters, outperforming uncalibrated 3GPP indoor-hotspot baselines. A complementary approach in [9] introduces a multi-modal neural radio radiance field that combines reference signal received power (RSRP) measurements with LiDAR point clouds for localized statistical channel modeling, reporting improved reconstruction accuracy and noise robustness over RSRP-only baselines. Outside the mmWave band, the PropEM-L framework [10] encodes raw point clouds for communication-aware multi-robot exploration and reports up to 44% improvement in received signal strength prediction over a log-distance baseline in subterranean environments. GPU-accelerated point-cloud ray tracing has also been demonstrated as a complementary technique for outdoor channel prediction [17]. The reported gains across these works are substantial, but the propagation regimes differ from the 2.4/5 GHz indoor industrial case: mmWave is dominated by direct-path attenuation and discrete blockage events, whereas industrial Wi-Fi is shaped by waveguiding through corridors and constructive/destructive multipath at length scales much larger than typical LiDAR feature scales. Whether geometric features that succeed in mmWave or in subterranean settings transfer to industrial Wi-Fi is therefore an open empirical question.

Dedicated LiDAR/Wi-Fi sensor-fusion work in the AGV domain has so far targeted localization, SLAM, mapping, and navigation drift compensation [18], [19], not link quality prediction. The above literature review reveals a gap at the intersection of LiDAR-derived environmental features, Wi-Fi link quality prediction, and industrial AGV deployment. The use-case study presented in this paper aims to address that gap by providing the experimental controls necessary to support a credible deployment-relevant conclusion.

III. Dataset Collection and Preparation

The experimental platform is a single industrial AGV equipped with a front-facing Leuze RSL 400 2D safety laser scanner (270° nominal angular range, operated at 0.2° angular resolution, ∼25 Hz scan rate) and a Navitrol on-board navigation and supervision system (NNS) by Navitec Systems. The facility is served by a single Wi-Fi AP whose coordinates were measured in the AGV's reference frame. Three data-gathering sessions were collected on a deployed AGV. The sessions cover three distinct routes, summarized in Table I.

TABLE I
Three-session dataset overview

Session	n_{pairs}	dur. (h)	f_{tel} (Hz)	$\overline{P}$ (dBm)	σ_P (dB)	motion %
S1	280,025	3.19	31	−36.91	10.52	37.6
S2	168,940	1.93	31	−39.66	7.35	39.3
S3	232,628	2.75	4.4	−31.07	6.22	61.3
total	681,593	7.87	—	—	—	—

f_{tel} is the on-board telemetry stack rate. "Motion %" is the fraction of recorded time during which the AGV is motion-active ($|\texttt{speed_mps}| > 0.1\,\text{m/s}$).

S1 is the largest session, both in duration and in spatial coverage, and is used for the within-session experiments. S1 and S2 record the on-board telemetry stack at 31 Hz; S3 at the slower 4.4 Hz, which is addressed by cadence-matching in the cross-session experiment (Section V). The mean per-session `signal_power` ranges from −39.66 to −31.07 dBm

[1]Dataset, source code, analysis scripts, and reports: https://github.com/Myroslav437/FLICS2026-LiDAR-Derived-WiFi-Quality-Prediction.

and varies most strongly within S1 ($\sigma_P = 10.52\,\mathrm{dB}$). The persistence of high- and low-signal regions along each route is visible in Fig. 1. The within-session contrast is sufficiently large that any deployment-grade predictor must resolve well below it.

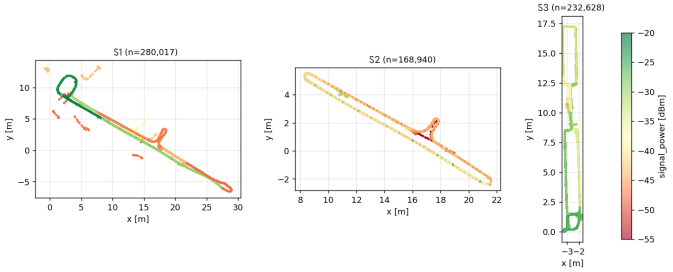


Fig. 1. AGV trajectory colored by `signal_power`. S1 and S2 share a common coordinate reference frame—equal coordinates correspond to the same physical location across the two sessions. S3 was mapped against an independently aligned frame.

A. Time synchronization

LiDAR scans and telemetry samples originate from independent hardware and software stacks with their own clocks. Even moderate temporal drift between the two streams biases any subsequent learning, because LiDAR-derived features would describe a slightly different position from the one the telemetry suggests. Time alignment is, therefore, a prerequisite for the use-case study and is performed before any feature engineering. A per-session offset $\hat{\tau}$ is estimated as the value that maximizes the cross-correlation between a telemetry signature and a LiDAR-side scan-to-scan dissimilarity signal, computed on a common 50 Hz grid with parabolic peak refinement. The estimation is performed once per session, and the resulting confidence interval is quantified via the moving-block bootstrap. Fig. 2 shows the cross-correlation curve for one session and the estimated peak.

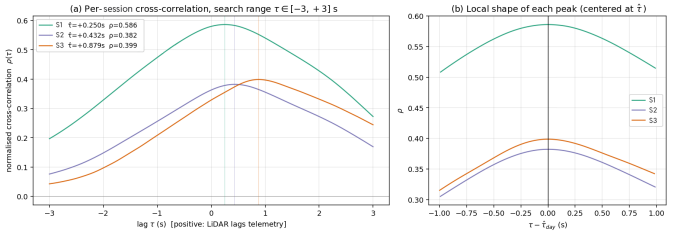


Fig. 2. Time synchronization between LiDAR and telemetry streams: (a) normalized cross-correlation between the telemetry signature and the LiDAR scan-to-scan dissimilarity signal; (b) local shape of each peak, centered at the corresponding $\hat{\tau}$. Positive τ corresponds to LiDAR lagging telemetry.

The hypothesis that a single offset applies to all three sessions is tested against the per-session estimates with Cochran's Q test ($Q = 8.06$, $df = 2$, $p = 0.018$); the test rejects the equal-offsets hypothesis, so per-session offsets are used. Within each session, two complementary stability tests support the use of a single constant offset. A half-split re-estimation, in which $\hat{\tau}$ is recomputed independently on the first and second half of each session, agrees within 95% confidence intervals (CIs) on all three sessions, with a maximum half-to-half $|\Delta\hat{\tau}| = 0.178\,\mathrm{s}$ on S1. A sliding-window drift fit, in which $\hat{\tau}$ is recomputed on overlapping 5–10 min windows and regressed against window-center wall-clock time, yields per-session slopes between -1.4 and $-0.6\,\mathrm{ms/min}$ ($p \geq 0.075$ on every session), with the accumulated within-session drift bounded by each session's bootstrap σ_{RMSE} on $\hat{\tau}$.

The estimated offsets, applied as constant per-session shifts to align the two streams, are $\hat{\tau}_{S1} = +0.250\,\mathrm{s}$, $\hat{\tau}_{S2} = +0.432\,\mathrm{s}$, and $\hat{\tau}_{S3} = +0.879\,\mathrm{s}$. After alignment and joining, 97.0% of LiDAR scans (681,593 of 702,624) have a successfully matched telemetry sample and are retained in the training-ready dataset.

B. Anomaly handling

The downstream models use the AGV's reported position (x_m, y_m) to compute every AP-relative feature. When on-board localization struggles—typically during initialization, after a manual repositioning of the AGV, in feature-poor environments, or under sensor degradation—the reported position becomes unreliable, and so do the features derived from it. Such rows must be excluded so that the feature derivation remains physically meaningful.

The NNS exposes a position-confidence score `nns_position_confidence` on a 0–100 scale, with a high value indicating that NNS estimates its own pose to be reliable. The distribution of this score across the entire dataset is provided in Fig. 3.

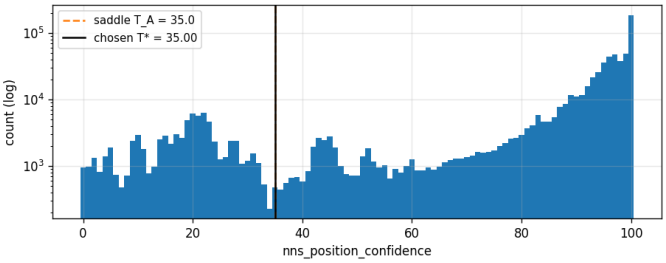


Fig. 3. Distribution of `nns_position_confidence` and the exclusion threshold $T^\star = 35$.

A 5-bin moving average of the histogram identifies a clear saddle between the two modes at $T^\star = 35$. Rows with confidence below this threshold are excluded from training and evaluation. The threshold removes 74,437 of the 681,593 matched samples (10.92%) and leaves the remaining 89.08% for downstream use.

C. Dataset noise floor

Reported model accuracy values are interpretable only against a reference for the irreducible variability of the data. We therefore characterize the dataset noise floor at the spatial resolution at which features are realistically deployable, namely 0.5 m position binning: if two visits to the same 0.5-m cell at different times yield substantially different `signal_power` readings, no model that uses 0.5-m position binning alone can resolve below that variability.

TABLE II
FEATURE STACK SUPPLIED TO THE PREDICTIVE MODELS, PARTITIONED INTO FIVE GROUPS

Group	#	Feature(s)	Type / units	Description
Position	2	x_m, y_m	float / m	AGV centroid in the workspace reference frame
Telemetry	1	`speed_mps`	float / $\mathrm{m\,s^{-1}}$	AGV linear speed
	1	`turn_rate`	float / $\mathrm{rad\,s^{-1}}$	AGV angular velocity
	1	`momentary_current`	float / A	instantaneous current consumption
AP-relative	1	`dist_to_AP`	float / m	Euclidean distance from AGV to surveyed AP
	2	`sin_AP`, `cos_AP`	float	sine and cosine of the AP bearing relative to the AGV heading
	1	`clutter_frac_toward_AP`	float / [0,1]	LiDAR clutter fraction in the 30° sector facing the AP (NaN if AP outside FOV)
	1	`is_AP_in_FOV`	binary	whether the AP bearing falls inside the LiDAR's empirical valid FOV
LiDAR scalar	2	`mean_dist_mm`, `dist_p90_mm`	float / mm	mean and 90th-percentile beam distance over the valid 222° sector
	2	`clutter_frac`, `openness_frac`	float / [0,1]	clutter and openness ratios over the valid 222° sector
	1	`mean_front_mm`	float / mm	mean beam distance in the front cone
LiDAR sectoral	7	`mean_dist_sector_{1..7}_mm`	float / mm	mean beam distance in each of seven contiguous 30° sectors
	7	`clutter_frac_sector_{1..7}`	float / [0,1]	clutter fraction in each of seven contiguous 30° sectors

The noise-floor estimate uses the 85 cells of size 0.5 m that contain at least 30 rows in both S1 and S2 (the two sessions sharing the same coordinate reference frame, allowing same-cell repeat measurements). The mean per-cell standard deviation of `signal_power` within a session is 4.90 dB on S1, 5.01 dB on S2, and 4.95 dB pooled. The median absolute difference between the two sessions' per-cell mean values is 4.24 dB. We therefore take 4.95 dB as the noise floor at 0.5-m position binning. Any predictor that resolves below this floor must be exploiting information at finer spatial granularity than 0.5 m — for instance, position at sensor resolution, telemetry, or AP-relative geometry.

D. Feature stack

The training-ready dataset comprises 29 numeric features, organized into five feature groups, as summarized in Table II. The empirical valid LiDAR field of view (FOV) is narrower than the nominal 270°. Of the 1,350 active beams, 241 hit the AGV body itself at sub-200 mm range and are permanently masked. The remaining 1,109 beams cover a contiguous 222° sector that hosts the seven 30° sectoral aggregates.

The experimental AGV is equipped with a single front-facing LiDAR, whereas production industrial AGVs typically use two for full 360° coverage. To account for this reduced angular coverage, evaluation metrics are stratified by `is_AP_in_FOV`. The in-FOV stratum is where directional LiDAR features are most informative: the AP falls within the sensor's valid sector, and obstruction measurements toward the AP are geometrically well-defined. This stratum also serves as the dual-LiDAR-equivalent lower bound — if LiDAR-derived features contribute predictive value, it should be observable here.

IV. METHODOLOGY

Two evaluation protocols are executed. The cross-session leave-one-route-out (LORO) protocol uses three folds, each holding out one session. The held-out session contains trajectory patterns the model has not been trained on. Because the coordinate reference frames do not align across all routes, cross-session models do not use the raw position (x_m, y_m); features are restricted to ego-frame LiDAR, telemetry, and AP-relative geometry.

The within-session LRO protocol uses five folds on session S1 only — the largest session by both spatial coverage and sample count. Regions are obtained by K-means clustering on (x_m, y_m) with $k = 5$. Within-session models do use the raw position. The two protocols together test the model's ability to transfer learned propagation structure across operating conditions (LORO) and to extrapolate to unseen physical regions of the same workspace (LRO).

The predictive model on both protocols is a gradient-boosted regression tree, trained with XGBoost (six hyperparameter configurations in total, defined below) and cross-checked with LightGBM. The target variable is the Wi-Fi received signal strength, in dBm, reported for the matched LiDAR–telemetry sample.

A. Feature group ablation

A single ablation between a full and an ablated model is ambiguous: no degradation can mean either (i) the removed feature group did not help, or (ii) the group is substitutable for another retained group. We therefore run two complementary ablations against the full model: M' removes feature group A (here, the LiDAR features) and M'' removes feature group B (here, the AP-relative features). The pair of resulting RMSE differences is then classified, with an absolute threshold of 0.5 dB, into one of four verdicts: *A removable*, *B removable*, *both removable*, or *mixed/unclear*. The procedure is feature group-agnostic: it reports which group can be removed without loss, independently of which group the analyst expected to matter. Cross-session variants are denoted B5, B5′, and B5″; within-session variants are denoted W4, W4′, and W4″.

B. Robustness controls

The validity of the ablation-based comparison depends on several conditions that are tested explicitly. In what follows,

TABLE III
CROSS-SESSION ABLATION VARIANTS

Variant	# feat.	Feature groups
B0	1	`dist_to_AP` only
B1	3	AP-relative geometry (no clutter)
B2	8	B1 + LiDAR scalar
B3	22	B2 + LiDAR sectoral
B4	25	B3 + telemetry
B5	27	full ego-frame LiDAR + AP-relative
B5′	8	B5 minus LiDAR features
B5″	22	B5 minus AP-relative features

the primary configuration of the XGBoost model is denoted H0 and is defined by tree depth 6, learning rate $\eta = 0.05$, L_2 regularization $\lambda = 1.0$, $n_{\text{est}} = 2000$ boosting rounds with early-stopping after 100 non-improving rounds, and random seed 20260427. Validation is performed chronologically per session (last 10 % of each training session). The robustness controls, applied to the relevant fold for each diagnostic, are:

1) *Hyperparameter sensitivity.* A six-configuration sweep of the XGBoost model: H0 (the primary configuration above); H1 (depth 4, otherwise H0); H2 (slow learning rate $\eta = 0.02$, higher regularization $\lambda = 5$); H3 (depth 8, otherwise H0); plus H1_{rv} and H2_{rv} with random rather than chronological validation splits.
2) *Framework sensitivity.* A LightGBM cross-check with default and depth-equivalent hyperparameters, addressing whether the comparison is specific to XGBoost.
3) *Feature informativeness check.* A LiDAR-feature permutation control: a per-session block-shuffle of the 19 ego-frame LiDAR columns that preserves their marginal and joint distributions but destroys row-level alignment with the target, evaluated both cross-session and within-session. If the shuffled-LiDAR model achieves the same ablation effect as the real-LiDAR model, the model was not extracting row-aligned information from the LiDAR features.
4) *Single-fold robustness.* For any within-session fold that nominally clears the threshold: a 1-m spatial buffer-zone exclusion (training rows within 1 m of any test row are dropped) controlling spatial autocorrelation, and a depth-4 hyperparameter swap controlling tree-depth overfitting; plus their joint application.
5) *Noise-floor reference.* The noise-floor characterization of Section III-C is used to contextualize the absolute RMSE values reported in Sections V and VI.

All training-set sizes after anomaly filtering are reported in Sections V and VI. Bootstrap 95 % CIs on RMSE are computed with $B = 1000$ resamples.

V. CROSS-SESSION EXPERIMENT

The cross-session experiment uses three folds: F1 (S1 held out, train on S2+S3), F2 (S2 held out, train on S1+S3), and F3 (S3 held out, train on S1+S2; the 31 Hz training sessions are decimated by $k = 7$ to match S3's 4.4 Hz cadence). Per-fold post-anomaly training set sizes are 337,390 rows for F1, 403,876 for F2, and 50,232 for F3. Six ablation variants B0–B5 progress from a distance-only baseline up to the full ego-frame LiDAR + AP-relative model (Table III). Two complementary variants implement the feature group ablation procedure: B5′ removes the LiDAR features while keeping AP-relative geometry, and B5″ removes the AP-relative features while keeping LiDAR.

Per-variant in-FOV RMSE on the cross-session folds under the H0 configuration is illustrated in Fig. 4. The highlighted headline variants B1 and B5 are considered representative of the full variant ladder. The intermediate variants B0/B2/B3/B4 represent graded enrichments, with per-fold per-variant values provided in the project repository (footnote 1) due to conference page constraints.

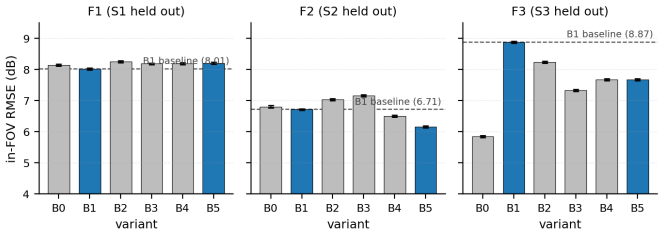


Fig. 4. Per-variant in-FOV RMSE on the cross-session folds. The dashed line marks RMSE(B1) per panel. Error bars are bootstrap 95 % CIs ($B = 1000$).

Table IV reports the in-FOV $\Delta_{\text{LiDAR}} = \text{RMSE}(B1) - \text{RMSE}(B5)$. We adopt $+1$ dB as a practical-relevance threshold (large enough to be measurable above bootstrap CIs, and roughly the magnitude at which an additional sensor stream could justify its integration cost in a deployment). The threshold is approached on only one of the three folds (F3, $+1.20$ dB); F1 and F2 are below it (-0.19 dB and $+0.55$ dB).

TABLE IV
CROSS-SESSION IN-FOV Δ_{LiDAR} PER FOLD UNDER H0

Fold	n_{test}	RMSE(B1)	RMSE(B5)	Δ_{LiDAR} (95 % CI)
F1	134,692	8.01	8.20	-0.19 ($[-0.21, -0.16]$)
F2	102,354	6.71	6.16	$+0.55$ ($[+0.54, +0.56]$)
F3	121,831	8.87	7.67	$\mathbf{+1.20}$ ($[+1.18, +1.23]$)

$\Delta_{\text{LiDAR}} > 0$ means LiDAR features improve over the baseline. RMSE values in dB.

The verdict of the feature group ablation procedure is summarized per fold in Table V. AP-relative features are not removable on any fold — B5″ is constantly worse than B5 throughout. LiDAR is removable on F1 and F2 (B5′ is within 0.5 dB of B5 in-FOV); on F3, B5 is below B5′ by ~ 0.5 dB on the in-FOV stratum, so the F3 verdict is "LiDAR complementary" rather than removable, motivating the F3-focused robustness diagnostic in the next subsection. Since F3 holds out S3, whose coordinate reference frame is aligned differently from the training sessions', the LiDAR feature distributions differ between training and test by construction.

TABLE V
CROSS-SESSION FEATURE GROUP ABLATION VERDICT PER FOLD (H0). IN-FOV RMSE IN DB

Fold	B5	B5′	B5″	verdict
F1	8.20	7.93	12.14	LiDAR removable
F2	6.16	5.85	7.51	LiDAR removable
F3	7.67	8.15	8.82	**LiDAR complementary**

A. F3 robustness diagnostic

We apply three robustness controls of Section IV-B to F3 to test whether the performance improvement reflects a deployment-relevant LiDAR signal (Table VI). In all six XGBoost and LightGBM configurations of Table VI, Δ_{LiDAR} stays in $[+1.11, +2.29]\,\mathrm{dB}$ and never crosses zero, so the headline is hyperparameter-stable and not framework-specific.

TABLE VI
F3 ROBUSTNESS DIAGNOSTIC (IN-FOV)

Diagnostic	Configuration	in-FOV Δ_{LiDAR} (dB)
Hyperparameter	H0 (depth-6, locked)	+1.20
Hyperparameter	H1 (depth-4)	+1.11
Hyperparameter	H2 (slow-η, high λ)	+1.51
Hyperparameter	H3 (depth-8)	+2.29
Framework	LightGBM, default	+1.13
Framework	LightGBM, n_{leaves}=15	+1.26
Permutation	block-shuffle, H0	+0.74
Permutation	block-shuffle, H1	+0.43

The permutation control, however, gives $\Delta^{\mathrm{shuffled}}_{\mathrm{LiDAR}} = +0.74\,\mathrm{dB}$ (H0) and $+0.43\,\mathrm{dB}$ (H1) — both strictly positive. This indicates that the $+1.20\,\mathrm{dB}$ on F3 is largely driven by the marginal-distribution difference between the independently aligned held-out coordinate reference frame and the training sessions' frame, rather than by row-aligned LiDAR information about the local environment. The F3 contribution, therefore, cannot be claimed as a deployment-relevant positive based on this evidence alone and is flagged as a limitation.

VI. WITHIN-SESSION EXPERIMENT

The within-session experiment evaluates whether LiDAR features add value when raw position is available — the regime in which a deployed model would normally operate inside a fixed workspace. LRO evaluation is performed on session S1 only, since S1 has by far the largest spatial coverage and sample count of the three sessions. The resulting partition is shown in Fig. 5.

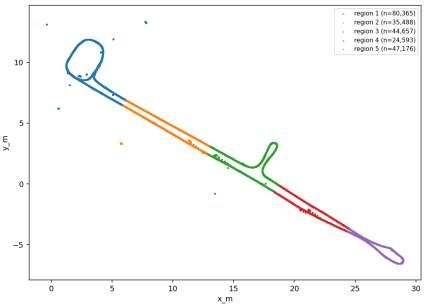


Fig. 5. Within-session spatial regions on S1.

The regions are obtained by K-means clustering on (x_m, y_m) with $k = 5$, and ordered by mean distance to the AP (R-1 closest, R-5 farthest). Each fold trains the predictive models on four regions and evaluates on the held-out region. Six ablation variants are evaluated as summarized in Table VII. W2 is the LiDAR-free baseline, and W4 is the full model; W4″ removes AP-relative features and is the counterpart of W2 in the feature group ablation procedure.

TABLE VII
WITHIN-SESSION ABLATION VARIANTS

Variant	# feat.	Feature groups
W0	2	position only
W1	5	W0 + telemetry
W2 (≡ W4′)	10	W1 + AP-relative (no LiDAR)
W3	15	W2 + LiDAR scalar
W4	29	W3 + LiDAR sectoral (full model)
W4″	24	W4 minus AP-relative

Per-variant in-FOV RMSE on the within-session folds under the H0 configuration is illustrated in Fig. 6. The highlighted headline variants W2 and W4 are considered representative of the variant ladder. The intermediate variants W0/W1/W3 are considered graded enrichments. Per-fold per-variant values are provided in the project repository (footnote 1) due to conference page constraints.

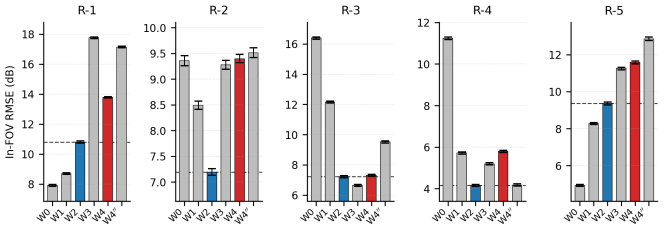


Fig. 6. Per-variant in-FOV RMSE on the within-session folds. The dashed line marks RMSE(W2) per panel. Error bars are bootstrap 95 % CIs ($B = 1000$).

The per-fold in-FOV $\Delta^{\mathrm{wi}}_{\mathrm{LiDAR}} = \mathrm{RMSE(W2)} - \mathrm{RMSE(W4)}$ values are reported in Table VIII. Every per-fold demonstrates $\Delta^{\mathrm{wi}}_{\mathrm{LiDAR}} \leq 0$, ranging from $-2.99\,\mathrm{dB}$ (R-1) to $-0.09\,\mathrm{dB}$ (R-3). The two largest negative values fall on the geometrically extremal folds R-1 and R-5 (closest and farthest from the AP), while the interior fold R-3 is closest to zero. This pattern suggests overfitting at the held-out extremes, though not conclusively established from the present data.

TABLE VIII
WITHIN-SESSION IN-FOV $\Delta^{\mathrm{wi}}_{\mathrm{LiDAR}}$ PER FOLD UNDER H0

Fold	n_{test}	RMSE(W2)	RMSE(W4)	$\Delta^{\mathrm{wi}}_{\mathrm{LiDAR}}$
R-1	57,270	10.80	13.79	−2.99
R-2	15,516	7.19	9.40	−2.20
R-3	30,974	7.22	7.31	−0.09
R-4	10,687	4.16	5.79	−1.63
R-5	20,245	9.37	11.58	−2.22

$\Delta^{\mathrm{wi}}_{\mathrm{LiDAR}} > 0$ means LiDAR features improve over the baseline. RMSE values in dB.

The feature group ablation verdict per fold (Table IX) is heterogeneous — but no fold yields a verdict in which LiDAR uniquely contributes. On every fold, removing LiDAR either improves RMSE or leaves it unchanged within tolerance. The robustness controls in the next subsections investigate the alternative interpretation that the negative result reflects spatial-autocorrelation contamination rather than a genuine inability of LiDAR features to add information.

TABLE IX
WITHIN-SESSION FEATURE GROUP ABLATION VERDICT PER FOLD (H0). IN-FOV RMSE IN DB

Fold	W4	W4′	W4″	verdict
R-1	13.79	10.80	17.14	mixed/unclear
R-2	9.40	7.19	9.51	AP-relative removable
R-3	7.31	7.22	9.52	LiDAR removable
R-4	5.79	4.16	4.18	mixed/unclear
R-5	11.58	9.37	12.86	mixed/unclear

A. Spatial autocorrelation in absolute RMSE

Within-session RMSE values cannot be read as straightforward generalization estimates: `signal_power` is spatially autocorrelated, and the held-out region rows are typically only a few meters away from the nearest training rows. The S1 semivariogram (Fig. 7) has empirical range ≈ 20 m, while the K-means regions span 5–13 m on a side, so training rows up to $\sim$10 m from any held-out region row are spatially correlated with the test set.

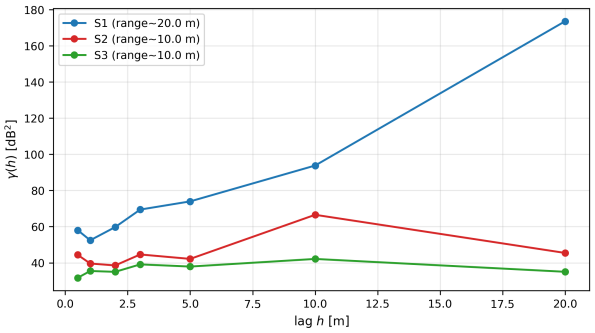


Fig. 7. Empirical semivariogram of `signal_power` per session. The lag h is the Euclidean distance between a pair of AGV (x_m, y_m) positions. The $\gamma(h) = \frac{1}{2N(h)} \sum_{(i,j)\in\mathcal{P}(h)} (s_i - s_j)^2$ is half the mean squared difference of `signal_power` over all row pairs whose distance falls in the bin $h \pm 0.25$.

To bound the magnitude of this contamination, we apply a **buffer-zone test** on R-1: dropping all training rows within 1 m of any held-out R-1 row inflates the in-FOV RMSE by $+10.21$ dB on W0, $+10.29$ dB on W1, $+5.63$ dB on W2, and $+13.86$ dB on W4. The contamination affects the LiDAR-free W2 baseline and the LiDAR-augmented W4 model differently within a fold, so the absolute RMSEs are fold-conditional rather than deployment-grade generalization estimates; the $\Delta^{\text{wi}}_{\text{LiDAR}}$ contrast is, however, taken between models trained and tested on the same row partition, so it is not inflated by spatial autocorrelation alone. The buffer-zone correction on R-4 (Table X) shifts $\Delta^{\text{wi}}_{\text{LiDAR}}$ from -1.63 dB (no buffer) to -0.01 dB (1-m buffer): the contrast moves toward zero, never above it, under the deployment-grade buffer correction.

B. R-4 robustness diagnostic

R-4 is the smallest fold in row count, which means that it is the most exposed to spatial autocorrelation and is the fold where any latent LiDAR contribution would be most likely to surface under a relaxed-overfitting hyperparameter swap. Table X reports R-4's in-FOV $\Delta^{\text{wi}}_{\text{LiDAR}}$ under diagnostic corrections and the LiDAR-feature permutation control.

TABLE X
R-4 ROBUSTNESS AND LIDAR PERMUTATION DIAGNOSTIC (IN-FOV)

Correction	probes	$\Delta^{\text{wi}}_{\text{LiDAR}}$ (dB)
none (H0 main run)	—	-1.63
1-m buffer	spatial autocorrelation	-0.01
depth-4 (H1)	tree-depth overfitting	$+0.60$
depth-4 + buffer	both, simultaneously	-0.64
LiDAR permutation (H0)	row-aligned vs. marginal	$+0.24$

Across the four corrections, $\Delta^{\text{wi}}_{\text{LiDAR}}$ stays below the $+1$ dB threshold throughout, and only the H1 single-correction exceeds zero. The within-session LiDAR-feature permutation control retrains W4 on R-4 with the 19 ego-frame LiDAR columns block-shuffled within session, preserving their marginal and joint distributions but destroying row-level alignment with the target. The shuffled-feature $\Delta^{\text{wi}}_{\text{LiDAR}} = +0.24$ dB, while the matched real-LiDAR $\Delta^{\text{wi}}_{\text{LiDAR}} = -0.01$ dB on the same training/validation partition. The shuffled fit achieves a marginally larger nominal contribution than the real fit. R-4's H0 fit therefore does not contain row-aligned LiDAR information beyond the marginal distributions of the LiDAR features. Combined with the four-correction diagnostic, R-4 supports the within-session negative result: no configuration produces a $\Delta^{\text{wi}}_{\text{LiDAR}}$ that is both above the $+1$ dB threshold and survives a deployment-relevant correction.

The combination of corrections that produces the most defensible deployment-relevant estimate is the LiDAR-free W2 model with the depth-4 hyperparameters and R-4 held-out region, producing an RMSE of 3.96 dB on `signal_power` in the in-FOV stratum. The 10 features of W2 are AGV position (x_m, y_m), three telemetry features, and five AP-relative features. The full 29-feature W4 model on the same fold and stratum achieves an RMSE of 3.36 dB, nominally 0.60 dB lower (consistent with the H1-no-buffer row of Table X); the gap does not survive the 1-m buffer correction. Both numbers sit approximately 1 dB below the 0.5-m position-binning noise floor of 4.95 dB (Section III-C), so the model's predictive power is exclusively explained by sub-cell information (position at sensor resolution, telemetry, and AP-relative geometry) rather than by coarse spatial binning. LiDAR features cannot contribute robust deployment-relevant information beyond what the 10-feature stack already encodes.

VII. FEATURE ATTRIBUTION ANALYSIS

We complement the ablation-based analysis with a model-internal view using TreeSHAP. The cross-session B5 fits and

the within-session R-3 W4 fit are analyzed. The cumulative $|\mathrm{SHAP}|$ over the features in each group is dominated by the AP-relative group on every fit (Fig. 8). On the cross-session folds, AP-relative contributes 6.7–7.7 in cumulative $|\mathrm{SHAP}|$ across 5 features (per-feature 1.3–1.5), whereas the LiDAR sectoral group contributes 1.4–2.0 across 14 features (per-feature 0.10–0.14). On the within-session R-3 W4 fit the AP-relative group cumulative is 6.3 over 5 features (per-feature 1.26) against 2.8 over 14 for LiDAR sectoral (per-feature 0.20). The ordering is consistent with the ablation evidence in Sections V and VI.

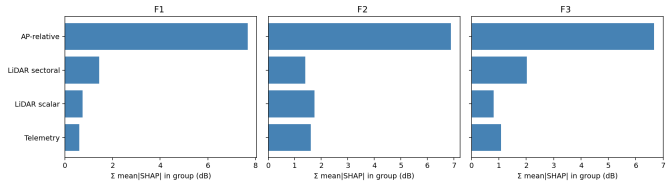


Fig. 8. SHAP feature group importance per cross-session fold (cumulative $|\mathrm{SHAP}|$ in dB).

We compute the Spearman correlation between each feature's value and its SHAP value, per cross-session fold, and flag a feature as sign-consistent if $|\rho| > 0.05$ with the same sign on all three cross-session folds. Of the four continuous AP-relative features tested, three pass (`dist_to_AP`, `sin_AP`, `cos_AP`); `clutter_frac_toward_AP` fails (the F2 fold gives an opposite sign to F1 and F3). The binary `is_AP_in_FOV` is excluded from the rank test. Among the 19 LiDAR features (5 scalar + 14 sectoral), 5 pass (`mean_dist_mm`, `clutter_frac`, `mean_dist_sector_3_mm`, `mean_dist_sector_6_mm`, `clutter_frac_sector_4`). Among the 3 telemetry features, 1 passes (`speed_mps`). The asymmetry — $3/4$ AP-relative against $5/19$ LiDAR — is informative: AP-relative features encode a transferable physical relationship that the gradient-boosted model picks up consistently across folds, whereas the LiDAR-derived environmental geometry does not, on this dataset. Plotting the per-cell dominant feature group in (x_m, y_m) shows that AP-relative and position together cover most of the AGV trajectory on every fold; LiDAR sectoral never dominates a contiguous spatial region. This is the per-cell counterpart of the group-importance result above.

VIII. Discussion

The results on the gathered dataset indicate that the dominant propagation effects in the workspace operate over distances substantially larger than the LiDAR's sensitivity range and are not localized to the AGV's immediate surroundings. The negative result is consistent with the fact that industrial Wi-Fi propagation is governed by waveguiding through corridors and constructive/destructive multipath at scales much larger than typical LiDAR feature scales [2]. The SHAP attribution analysis (Section VII) supports this reading: AP-relative features dominate the model's predictions across every fold, and the per-feature contribution of LiDAR sectoral features is the smallest of any group.

The 10-feature stack (x_m, y_m + 3 telemetry features + 5 AP-relative features) achieves an RMSE of 3.96 dB on `signal_power` in the within-session LRO evaluation under depth-4 hyperparameters — approximately 1 dB below the 0.5-m position-binning noise floor of 4.95 dB. LiDAR-derived environment features did not contribute to either cross-session generalization (LiDAR removable on the F1 and F2 folds; F3's nominal positive diagnosed as marginal-distribution-driven) or within-session generalization under proper testing (no fold's $\Delta^{\mathrm{wi}}_{\mathrm{LiDAR}}$ both cleared the threshold and survived a deployment-relevant correction). For the studied deployment scenario, the LiDAR sensor-fusion engineering effort did not add value over the 10-feature AP-aware baseline.

The finding is conditional on the experimental design: a closed-loop trajectory, a single fixed AP per session, and summary-statistic LiDAR features. Under these conditions (x_m, y_m) and AP-relative geometry are highly informative on their own, leaving little residual variance for LiDAR-derived features — which are themselves functions of position — to explain. The result is limited to the utilized deployment and is not a universal statement about LiDAR–Wi-Fi fusion. A further conditioning factor is the LiDAR feature representation: the 19 features used here are hand-designed summary statistics, not learned point-cloud encoders such as the PointNet-based regressor of [8]. Whether such richer representations would change the present conclusions is an open empirical question.

Several **limitations** bound the scope of this paper's claims:

- *Single facility, single AP, three sessions.* The experiments are conducted on one AGV in one facility, across three routes.
- *Single front LiDAR.* The AGV has a single front-facing 270° LiDAR (222° empirical valid sector), whereas production AGVs typically carry front+rear LiDARs. The in-FOV stratum used in the paper is the dual-LiDAR-equivalent, and the negative result holds in both strata. But this does not fully substitute for dual-LiDAR coverage, since a rear LiDAR observes geometry behind the AGV not represented in any forward-only feature.
- *Single independently aligned reference frame.* F3's nominal $+1$ dB Δ_{LiDAR} is diagnosed as marginal-distribution-driven (Section V-A). A definitive evaluation requires additional sessions on independently mapped routes.
- *Closed-loop trajectory.* All sessions follow predefined AGV routes traversed in a closed loop, with no free-roaming exploration.
- *AP fixed within each session.* Each session uses a single stationary AP, so AP-relative geometry alone captures most signal variance attributable to global geometry.
- *Spatial autocorrelation.* The S1 semivariogram range (≈ 20 m) is comparable to the within-session region size and a 1 m buffer does not fully decorrelate, so the absolute within-session RMSE values within a fold should be

read as fold-conditional rather than as deployment-grade generalization estimates.

Four experimental directions would relax the position-redundancy condition under which LiDAR-derived features cannot earn their keep: (i) *environmental reconfiguration* between repetitions of the same loop — moving pallets, opening or closing doors — holding (x_m, y_m) constant while changing multipath geometry; (ii) *AP placement varied across sessions* in the same map, decoupling signal from absolute location; (iii) *multi-AP infrastructures with simultaneous measurements*, isolating the directional contribution of LiDAR features per AP within a shared scan; (iv) *open-trajectory or space-filling exploration* in place of closed-loop traversal, reducing the spatial redundancy that makes position informative on its own.

IX. Conclusion

This paper presents a use-case study that investigates whether LiDAR-derived environmental geometry can improve Wi-Fi signal strength prediction for industrial AGVs over what AP-relative features alone achieve. The study used cross-session and within-session generalization protocols on 681,593 matched LiDAR-telemetry pairs from three data-gathering sessions covering three routes. Across the experimental configurations tested, ego-frame LiDAR features did not contribute robust value over a 10-feature baseline of position, telemetry, and AP-relative geometry derived from surveyed AP coordinates. A single nominal positive was diagnosed by a LiDAR-feature permutation control as marginal-distribution-driven rather than row-aligned LiDAR signal. The lowest-RMSE within-session LiDAR-free model achieves an RMSE of $3.96\,\mathrm{dB}$ on `signal_power`, approximately $1\,\mathrm{dB}$ below the dataset's 0.5-m position-binning noise floor. Future work includes cross-facility replication, dual-LiDAR coverage, additional sessions on independently aligned reference frames, and pose-uncertainty modeling.

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